# DATA SCIENCE LAB TASK

*Python Basics and Data Cleaning*

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## 1. Introduction

Python is a programming language used for many tasks, including data science. In this assignment, I have practiced basic Python concepts and cleaned a student dataset.

## 2. Short Theory

A library contains ready-made code. I used pandas to work with the dataset. A loop repeats instructions, and a list stores several values in one variable.

Data can be structured (tables or CSV files), unstructured (images, videos, text), or semi-structured (JSON and XML).

## 3. Basic Python Tasks

- Calculator: uses a function and if/elif conditions to do simple calculations.
- Loop and list: prints numbers greater than 25 from a list.
- Function: calculates the average of a list of numbers.

## 4. Data Cleaning Task

I created 100 student records with names, ages, cities, and marks. Some values were missing or incorrect, and some names had extra spaces or different capital letters. I also added a few duplicate rows.

Cleaning steps:

- Removed extra spaces and made names and cities consistent.
- Replaced invalid ages and marks with missing values, then filled them using the median.
- Filled missing cities with “Unknown”.
- Removed duplicate rows.
- Saved the cleaned data as cleaned_student_dataset.csv.

## 5. Python Code

The code below covers the basic tasks and data cleaning used in this assignment.

```
import pandas as pd

# Make a small student dataset with 100 entries
names = [" Ali ", "SARA", "Ahmed", " Ayesha", "Bilal ",
         "Hina", "USMAN", "zara", "Hamza", "Noor"]
cities = ["Lahore", " karachi ", "ISLAMABAD", "Peshawar", "lahore"]

records = []
for i in range(100):
    records.append({
        "Student_ID": i + 1,
        "Name": names[i % 10],
        "Age": [20, 21, None, 19, 22, 200, -5, 23, 21, None][i % 10],
        "City": cities[i % 5] if i % 13 != 0 else None,
        "Marks": [85, 92, 78, None, 88, 91, 75, 84, 101, 67][i % 10]
    })

df = pd.DataFrame(records)

# Add some duplicate rows
df = pd.concat([df, df.iloc[[4, 12, 24]]], ignore_index=True)
```

```
print("Original data:")
print(df.head())
print(df.isnull().sum())

# Clean text
df = df.replace(r'^\s*$', None, regex=True)
df["Name"] = df["Name"].astype("string").str.strip().str.title()
df["City"] = df["City"].astype("string").str.strip().str.title()

# Fix invalid ages and marks
df.loc[~df["Age"].between(0, 100), "Age"] = None
df["Age"] = df["Age"].fillna(df["Age"].median())

df.loc[~df["Marks"].between(0, 100), "Marks"] = None
df["Marks"] = df["Marks"].fillna(df["Marks"].median())

df["City"] = df["City"].fillna("Unknown")

# Remove duplicate rows
df = df.drop_duplicates().reset_index(drop=True)

print("\nCleaned data:")
print(df.head())
print(df.isnull().sum())

# Save cleaned data
df.to_csv("cleaned_student_dataset.csv", index=False)
print("File saved.")

# Calculator
def calculator(a, b, operation):
    if operation == "+":
        return a + b
    elif operation == "-":
        return a - b
    elif operation == "*":
        return a * b
    elif operation == "/":
        if b != 0:
            return a / b
        return "Cannot divide by zero"
    else:
        return "Invalid operation"

print(calculator(12, 3, "*"))

# List and loop
numbers = [10, 20, 30, 40, 50]
for num in numbers:
    if num > 25:
        print(num)

# Function
def average(values):
    return sum(values) / len(values)

print("Average:", average(numbers))
```

## 6. Result

The program shows the original data, checks missing values, cleans the dataset, displays a preview of the cleaned data, and saves it to a CSV file.

## 7. Conclusion

In this task, I practiced Python lists, loops, conditions, functions, and basic data cleaning using pandas.